

# BANK MARKETING ANALYTICS

## Campaign Analysis Report

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### 1. Introduction

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This report presents a comprehensive analysis of a bank marketing campaign dataset. The campaign was conducted by a financial institution to promote term deposit subscriptions among its existing customers through direct phone calls. The goal of this analysis is to understand customer behaviour, identify key drivers of subscription, and provide actionable recommendations to improve future campaign performance.

The analysis was performed using MySQL Workbench for data exploration and cleaning, and Power BI Desktop for interactive visualization and dashboard development.

### 2. Business Problem Statement

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The bank conducted a marketing campaign targeting existing customers via telephone. However, not all customers responded positively. The business raised two key questions:

- Why do some customers subscribe to a term deposit while others do not?
- Which customer groups are more likely to respond positively to marketing campaigns?

Answering these questions enables the bank to focus its resources on high-potential customer segments, reduce campaign costs, and improve overall conversion rates.

### 3. Dataset Description

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The dataset used in this analysis is the Bank Marketing Dataset, which contains records of direct marketing campaigns. Below is a summary of the dataset:

- Total Records: 11,162 customer entries
- Total Columns: 17 attributes plus 1 derived column (contacted before)
- Target Variable: deposit — indicates whether the customer subscribed to a term deposit (yes/no)

Key columns include:

- Age — Customer age
- Job — Type of employment
- Marital — Marital status
- Education — Level of education
- Balance — Average yearly account balance
- Contact — Communication type used
- Duration — Duration of last call in seconds
- Poutcome — Outcome of previous marketing campaign
- Deposit — Target variable (yes/no)

## 4. Data Analysis Process

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### 4.1 Data Import and Preparation (MySQL)

The raw dataset was imported into MySQL Workbench using the `LOAD DATA INFILE` command. A duplicate working table (`bank_2`) was created from the raw table (`bank_raw`) to preserve the original data before any cleaning was applied.

### 4.2 Missing Value Assessment

A thorough check for NULL and empty values was conducted across all 17 columns. The dataset did not contain traditional NULL values. Instead, missing information was encoded as the string 'unknown' in categorical columns such as job, education, contact, and poutcome. The pdays column used -1 to represent customers who had never been contacted in a previous campaign.

### 4.3 Feature Engineering

A new column, contacted before, was created to improve readability. Customers with pdays = -1 were labelled 'No' (never previously contacted), while all others were labelled 'Yes'. This made the data more interpretable for visualization in Power BI.

### 4.4 Exploratory Data Analysis (EDA)

Subscription rates were calculated and compared across multiple customer segments using SQL GROUP BY queries and conditional aggregation (CASE WHEN). The following factors were analysed:

- Age group distribution and subscription rate
- Job type and subscription likelihood
- Education level and subscription behaviour
- Contact method effectiveness
- Previous campaign outcome influence
- Marital status and subscription rate

### 4.5 Visualization (Power BI)

The cleaned dataset was loaded into Power BI Desktop. DAX measures were created for Total Subscribers, Subscription Rate, Average Balance, and Average Call Duration. An interactive dashboard was built featuring KPI cards, bar charts, donut charts, and a line chart for monthly trends.

## 5. Key Findings

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| KPI Metric                | Value              |
|---------------------------|--------------------|
| Total Customers Contacted | 11,162             |
| Total Subscribers         | 5,000 (approx. 5K) |
| Average Account Balance   | 1.53K              |
| Overall Subscription Rate | 47.38%             |
| Average Call Duration     | 371.99 seconds     |

### 5.1 Subscription by Age Group

Customers aged 60 and above had the highest subscription rate at 77%, followed by customers under 30 at 60%. The middle-aged groups (40–49) had the lowest subscription rate at 40%, suggesting older and younger customers are more receptive to term deposit campaigns.

### 5.2 Subscription by Job Type

Students (75%) and retired customers (66%) showed the highest subscription rates. Blue-collar workers had the lowest rate at 36%, indicating that job type is a strong predictor of campaign success.

### 5.3 Subscription by Education

Customers with tertiary education had the highest subscription rate at 28.64%, followed by secondary education at 26.83%. Customers with unknown education levels were the least likely to subscribe.

### 5.4 Subscription by Contact Method

Cellular contact was the most widely used method at 42.67% of all contacts. Telephone contact showed a higher unknown rate. Unknown contact methods had the lowest subscription conversion, reinforcing the importance of direct and identifiable communication channels.

### 5.5 Previous Campaign Contact

74.57% of customers had never been contacted in a previous campaign, while 25.43% had been contacted before. This shows the campaign largely targeted fresh customers with no prior campaign history.

### 5.6 Monthly Subscription Trend

The subscription rate showed a declining trend from the beginning of the year through to the later months, with the highest engagement observed in the earlier campaign months. This suggests that early campaign timing yields better results.

## 6. Business Insights

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- Older customers (60+) and students are the most valuable target segments due to their significantly higher subscription rates.
- Cellular phone contact is more effective than telephone or unknown contact methods and should be prioritized in future campaigns.
- Retired and student job groups show the strongest positive response to term deposit offers, likely due to financial stability and savings intent respectively.
- Tertiary-educated customers are more likely to subscribe, suggesting they better understand the value of term deposits.
- Customers who were previously contacted and had a successful outcome are far more likely to subscribe again, highlighting the importance of maintaining relationships with past positive responders.
- Early campaign months drive higher subscription rates, suggesting that the bank should front-load campaign efforts at the start of the year.

## 7. Recommendations

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- Target high-value segments: Focus marketing efforts on customers aged 60 and above, students, and retired individuals as these groups consistently show the highest subscription rates.
- Prioritize cellular communication: Since cellular contact yields the most engagement, the bank should ensure customer mobile numbers are captured and used as the primary contact channel.
- Re-engage previous successful contacts: Customers who responded positively in past campaigns should be prioritized in future campaigns as they carry a significantly higher conversion likelihood.
- Improve education-based targeting: Campaigns should be tailored with clearer, simpler messaging for customers with lower education levels to improve their conversion rates.
- Schedule campaigns early in the year: Given that subscription rates decline over the months, campaigns should be launched as early as possible to maximize conversion.
- Reduce contact with low-performing segments: Blue-collar and middle-aged customer groups show consistently low subscription rates. The bank should reduce spend on these segments or redesign its messaging strategy for them.

## 8. Conclusion

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This analysis successfully identified the key factors that influence whether a customer subscribes to a term deposit. By leveraging MySQL for structured data exploration and Power BI for interactive visualization, meaningful patterns were uncovered across age, job type, education, contact method, and campaign history.

The overall subscription rate of 47.38% indicates a fair effective campaign, but there is clear room for improvement through smarter targeting. By focusing resources on high-converting customer segments — particularly older customers, students, retirees, and those with tertiary education — and by prioritizing cellular contact and early campaign timing, the bank can significantly improve future campaign performance and return on investment.